package built around a chip, solder, and a gold-plated submount, for example, the submount alone could be swapped for a gold-metallised diamond version while the solder stays the same, making diamond, in some cases, a fairly simple drop-in replacement.

The hard part: growing diamonds big enough and pure enough

So, what makes a large piece of single-crystal diamond difficult to grow? The first hurdle is size: single-crystal diamond for electronics is currently produced in areas around 20 × 20mm, with 30 × 30mm near the practical limit today. Crystal quality gets harder to maintain as the size grows.

Then comes purity. For a material meant to move heat efficiently, impurities like nitrogen are unwelcome, since nitrogen atoms interfere with heat transport through the crystal. Boron and other impurities also need to be kept in check, depending on the application.

But how is that control achieved? In CVD growth, it starts with the gas mixture. Controlled amounts of gas can be introduced into the hydrogen-methane feed using mass flow controllers. But the outcome depends on much more than the gas composition alone. Substrate orientation, methane concentration, temperature, pressure, plasma conditions, and growth rate all shape the final crystal, and because these variables interact with one another, changing one can shift the effects of the rest.

That makes detailed characterisation essential after every growth run, turning the process into a feedback loop: grow the diamond, measure its properties, study the results, and adjust the growth conditions accordingly. AI-based analysis is increasingly being used to spot relationships between growth parameters and material properties, helping tighten that loop.

COMPANIES WORKING ON SCIENTIFIC APPLICATIONS OF DIAMONDS

Company	Focus
Diamfab	Diamond semiconductors, including diamond wafers, Schottky diodes, FETs, power electronics, and quantum applications.
Element Six	Part of De Beers Group, develops synthetic diamonds for industrial and high-tech applications, including thermal management, photonics, sensors, and quantum technologies.
Kara	Develops engineered single-crystal diamond for thermal management, power electronics, quantum sensing, photonics, quantum networks, and research applications.

Table B
Key point: *Many companies producing diamonds for industrial applications also serve the jewellery market. Companies focused mainly on scientific and technical uses therefore represent a more specialised segment.*

Where diamond is already proving itself

Space companies are exploring diamonds for systems with demanding thermal requirements. AI hardware is another major area. As GPUs and other AI chips become more powerful, heat is becoming concentrated in smaller areas, increasing the cooling burden in data centres.

While diamond can move heat away from the chip more efficiently than materials such as copper, the actual improvement depends on the system design, including the diamond's thermal conductivity, size, and position within the package. There is therefore no single temperature reduction that applies to every application.

Lower junction temperatures also buy devices more thermal headroom: a GPU can hold its performance without running as close to its thermal ceiling, which can ease thermal stress and potentially extend device lifetime. The same effect ripples out to data-centre cooling: when GPUs run cooler, cooling systems don't have to work as hard, which can reduce reliance on intensive liquid cooling and, potentially, the electricity spent on cooling overall.

For now, thermal management is the clearest semiconductor application for diamond, especially in systems where heat has become the dominant limit on performance.

Beyond cooling: diamond as an active semiconductor

The next frontier is diamond not just as a heat spreader, but as a semiconductor in its own right. Most current work focuses on placing diamond above or below chips to improve thermal management—important, but only part of the picture. The bigger opportunity may lie in developing diamond as an active semiconductor material. Research into reliable diamond-based semiconductor devices, and ways to manufacture them consistently, could push the material beyond cooling duty and into a larger role in electronics. EFY

Can India turn its Diamond Hub into a Tech Hub?

- ***Surat's advantage.*** Surat is a major global diamond hub, with around 90 per cent of the world's diamonds passing through the city's ecosystem at some stage.
- ***Beyond jewellery.*** Several Surat-based companies are now working on synthetic diamond applications for industrial and technology markets.
- ***Indian players.*** Anjali Semicon, SP3 Technologies, Vegasemi Tech, and JB Dianet are among the companies developing synthetic-diamond materials and products for applications beyond jewellery, including semiconductors, thermal management, and so on.
- ***Early opportunity.*** As these applications develop, Surat's existing diamond ecosystem could help India build a larger role in the emerging synthetic-diamond technology market.

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